BLG 317E - DATABASE SYSTEMS

FALL 2024 (CRN: 13591)

PROJECT PROPOSAL

Note that your selected dataset should be extensive enough to allow at least 5 main tables, and each table must have at least 3 non-key columns. Each team member must be responsible for at least one main table. Do your best in identifying table requirements. You will be asked to change your selected dataset if it does not seem to meet the minimum requirements. Please see the project evaluation and project self-assessment forms before finalizing your dataset to get an idea about the project evaluation criteria and process.

<table>
<tr><td>Project Group Name</td><td>Medico</td></tr>
<tr><td>Dataset title</td><td>Hospital Management Dataset</td></tr>
<tr><td>Dataset URL</td><td>https://github.com/MedicoDB/Medico_db/tree/main/Dataset</td></tr>
<tr><td>Data description</td><td>
1) patients.csv — Patient demographics & enrollment - MAIN

PK: patient_id

FK: insurance_type → insurers.code

non-key columns : first_name, last_name, dob, age, gender, ethnicity, insurance_type, marital_status, address, city, state, zip, phone, email, registration_date

Represents the FHIR Patient concept; demographics are stable reference data joined by patient_id.

2) providers.csv — Clinicians & identifiers - MAIN

PK: provider_id

FK: head _id → specialty_heads.head _id

Non-key: name, department, specialty, npi, inhouse, location, years_experience, contact_info, email

npi is the U.S. National Provider Identifier (10 digits), the standard identifier in HIPAA transactions.

3) encounters.csv — Visits/episodes of care - MAIN

PK: encounter_id

FK:
• patient_id → patients.patient_id
• provider_id → providers.provider_id
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	Non-key: visit_date, visit_type, department, reason_for_visit, diagnosis_code, admission_type, discharge_date, length_of_stay, status, readmitted_flag A FHIR Encounter: the actual interaction between a patient and provider(s). Use patient_id/provider_id to link. **4) diagnoses.csv — Diagnostic coding per encounter - MAIN** **PK:** diagnosis_id **FK:** encounter_id → **encounters.encounter_id** **Non-key:** diagnosis_code, diagnosis_description, primary_flag, chronic_flag Diagnosis_code is intended for ICD-10/ICD-10-CM coding of conditions; primary_flag denotes the principal diagnosis. **5) procedures.csv — Procedures/services rendered - MAIN** **PK:** procedure_id **FK:** • encounter_id → **encounters.encounter_id** • provider_id → **providers.provider_id** **Non-key:** procedure_code, procedure_description, procedure_date, procedure_cost Procedure_code is typically CPT (or HCPCS) for billing/reporting of services. **6) lab_tests.csv — Laboratory tests & results - MAIN** **PK:** lab_id **FK:** encounter_id → **encounters.encounter_id** **Non-key:** test_name, test_code, specimen_type, test_result, units, normal_range, test_date, status test_code is designed for LOINC, the standard coding for lab orders/results; each record is an individual lab observation. **7) medications.csv — Medications prescribed/administered - MAIN** **PK:** medication_id **FK:** • encounter_id → **encounters.encounter_id** • prescriber_id → **providers.provider_id**

	Non-key: drug_name, dosage, route, frequency, duration, prescribed_date, cost Captures medication orders/administrations tied to an encounter (maps to FHIR medication resources conceptually). **8) claims_and_billings.csv — Charges, claims, payments - MAIN** **PK:** billing_id **FK:** • patient_id → **patients.patient_id** • encounter_id → **encounters.encounter_id** • insurance_provider → **insurers.code** **Non-key:** insurance_provider, payment_method, claim_id, claim_billing_date, billed_amount, paid_amount, claim_status, denial_reason Represents charge/claim lines and adjudication status at a summary level; claim_id links to denial details. The FHIR Claim resource is the analog for reimbursement requests. **9) denials.csv — Denials & appeals lifecycle – MAIN** **PK:** denial_id **FK:** • claim_id → **claims_and_billing.claim_id** Columns: claim_id, denial_id, denial_reason_code, denial_reason_description, denied_amount, denial_date, appeal_filed, appeal_status, appeal_resolution_date, final_outcome. Tracks denial reasons and appeal outcomes per claim. Useful for payer performance and revenue cycle analysis. (Claim/denial processes align with FHIR financial resources.) **10) insurers.csv — Payer dictionary — EXTRA (lookup)** **PK:** insurer_id **Unique natural key:** code *(e.g., UHC, Aetna, BCBS, Humana, Cigna, Medicare, Medicaid)* **Non-key:** name, payer_type *(commercial/public)* **Referenced by:** patients.insurance_type → insurers.code **Notes:** Centralized codebook to standardize payer names/codes. **11) specialty_heads.csv — One head doctor per specialty — EXTRA (organizational/lookup)**

PK: head_id
FK (recommended & implemented): head_provider_id → providers.provider_id *(validates that the head is a real provider)*
Non-key: specialty, head_name, head_email
Referenced by: providers.head_id → specialty_heads.head_id
Notes: Ensures a single designated head per specialty for governance and reporting.

Every MAIN Table has at least 1 FK as T.A. told us to do, We have added 2 extra tables for patients FK and providers FK

Member 1 Signature	Full name	Abdulsamet Ekinci
	Student no	150220723
	Responsibility:	Managing team – DB Analysis Tables: patients (**FK:insurance_type**), **encounters (FK: patient_id, provider_id), insurers(PK)**
Member 2 Signature	Full name	Uğur Erkan Bol
	Student no	150210074
	Responsibility	UI/UX design – Tester Tables : providers (FK: head_id), diagnoses (FK: encounter_id), specialty_heads(FK: head_provider_id)
Member 3 Signature	Full name	Furkan İslamoğlu
	Student no	150220065
	Responsibility:	Frontend Lead Tables: **procedures (FK: encounter_id, provider_id)**, **medications (FK: encounter_id, prescriber_id)**
Member 3 Signature	Full name	Mustafa Talha Hamsoğlu
	Student no	150220066
	Responsibility:	Backend Lead

		Tables: **calims_and_billing (FK: patient_id, encounter_id)**, **denials (FK: claim_id → billing.claim_id)**
Member 4 Signature	Full name	Mustafa Yunus Diler
	Student no	150210003
	Responsibility:	Backend Optimization Tables: lab_tests (FK: encounter_id)